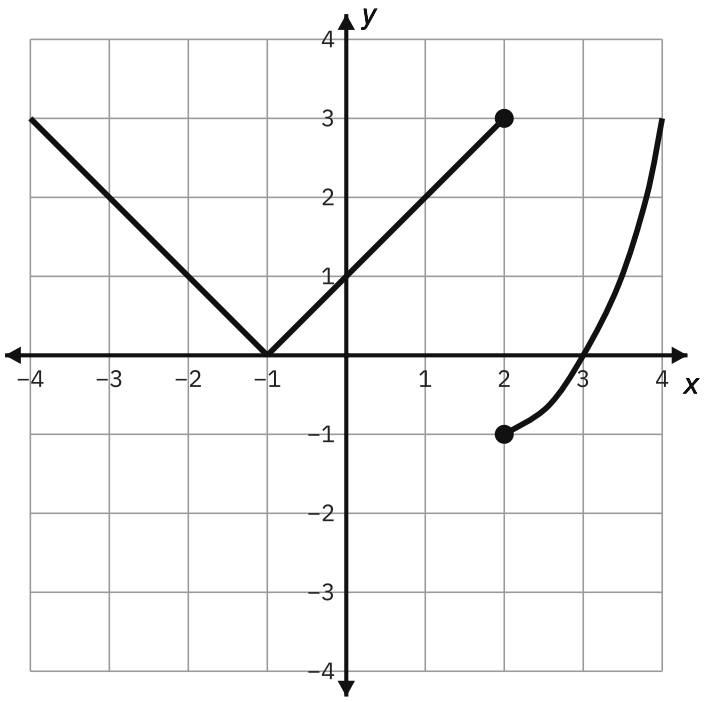
STATION

1

UNIT 5 REVIEW #2

# Is it a function?

Marcel claims that the graph below represents a function.




State whether Marcel is correct. Justify your answer.

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STATION

2

UNIT 5 REVIEW #2

# Which relation is a function?

Which relation is a function?

**1)** {(2, 1), (3, 1), (4, 1), (5, 1)}

**2)** {(1, 2), (1, 3), (1, 4), (1, 5)}

**3)** {(2, 3), (3, 2), (4, 2), (2, 4)}

**4)** {(1, 6), (2, 8), (3, 9), (3, 12)}

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STATION

3

UNIT 5 REVIEW #2

# Evaluate a function

If $f(x) = \dfrac{\sqrt{2x + 3}}{6x - 5}$, then $f(\dfrac{1}{2}) =$

**a)** 1

**b)** −2

**c)** −1

**d)** $-\dfrac{13}{3}$

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STATION

4

UNIT 5 REVIEW #2

# Evaluate a function

A function is defined as $K(x) = 2x^2 - 5x + 3$. The value of $K(-3)$ is

**a)** 54

**b)** 36

**c)** 0

**d)** −18

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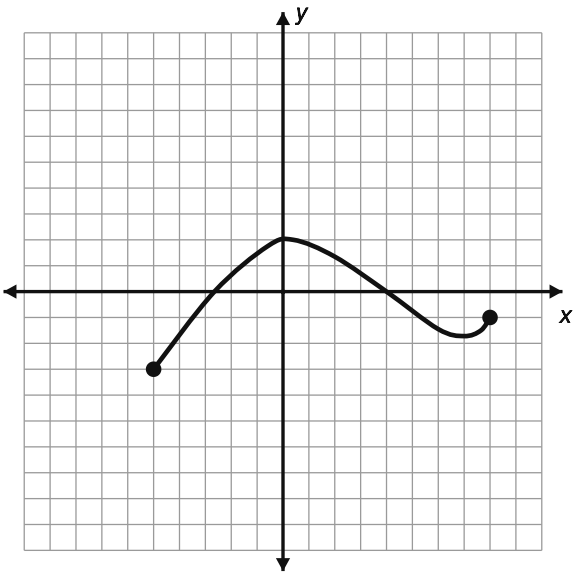
STATION

5

UNIT 5 REVIEW #2

# Reading a graph

The graph below represents the function $y = f(x)$.




**a)** State the domain and range of this function.

**A)** Domain: $-5 \le x \le 8$
Range: $-3 \le y \le -1$

**B)** Domain: $-5 \le x \le 8$
Range: $-3 \le y \le 2$

**C)** Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 8$

**D)** Domain: $-10 \le x \le 10$
Range: $-10 \le y \le 10$

**b)** What is the value of $x$ when $f(x) = -3$?

**c)** What is $f(4)$?

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STATION

6

UNIT 5 REVIEW #2

# Subtracting polynomials

What is $(x - 4)^2$ subtracted from $(x^2 + 3x - 4)$?

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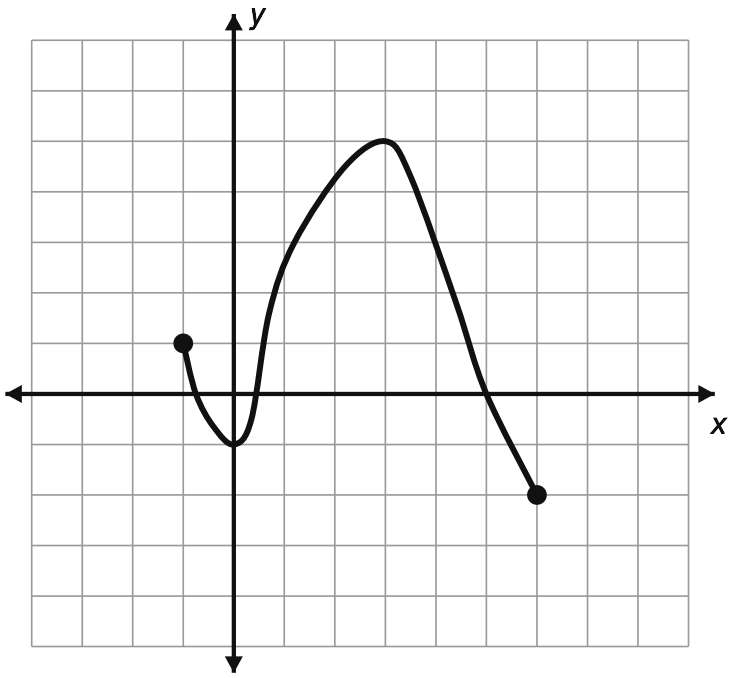
STATION

7

UNIT 5 REVIEW #2

# Domain from a graph

What is the domain of the function shown below?




**1)** $-1 \le x \le 6$

**2)** $-1 \le y \le 6$

**3)** $-2 \le x \le 5$

**4)** $-2 \le y \le 5$

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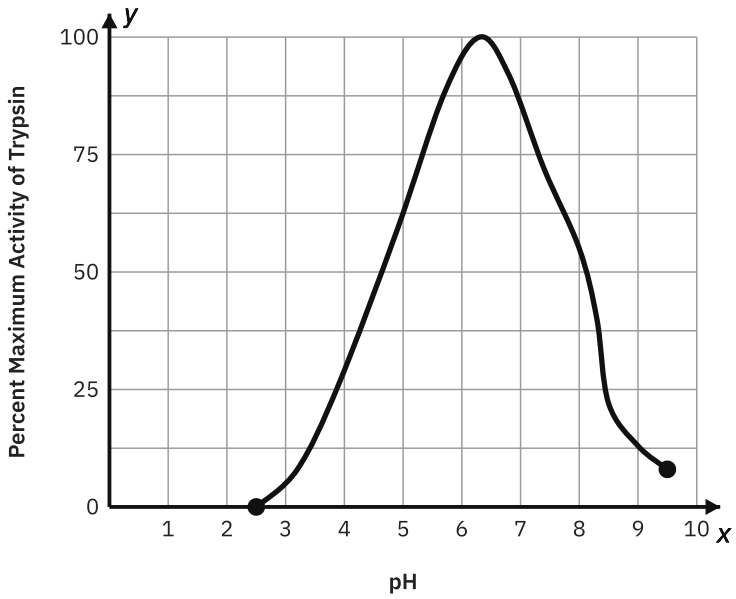
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8

UNIT 5 REVIEW #2

# Range of data

Data collected during an experiment are shown in the accompanying graph.




What is the range of this set of data?

**1)** $2.5 \leq y \leq 9.5$

**2)** $2.5 \leq x \leq 9.5$

**3)** $0 \leq y \leq 100$

**4)** $1 \leq x \leq 10$

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STATION

9

UNIT 5 REVIEW #2

# Terms from a formula

Find the first 5 terms given the formula $f(n) = 3(n - 1) + 4$

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STATION

10

UNIT 5 REVIEW #2

# Terms from a formula

Find the first 5 terms given the formula $f(n) = 2(5)^{n-1}$

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STATION

11

UNIT 5 REVIEW #2

# Sequence: 1, 2, 4, 8, ...

Given the following: 1, 2, 4, 8, 16, 32, ...

**a)** Is the pattern Arithmetic, Geometric, or Neither?

**A)** Arithmetic

**B)** Geometric

**C)** Neither

**b)** Write a rule that represents the *n*th term of the sequence.

**c)** Find the 10th term of the sequence.

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STATION

12

UNIT 5 REVIEW #2

# Sequence: 4, 7, 10, ...

Given the following: 4, 7, 10, 13, 16, ...

**a)** Is the pattern Arithmetic, Geometric, or Neither?

**A)** Arithmetic

**B)** Geometric

**C)** Neither

**b)** Write a rule that represents the *n*th term of the sequence.

**c)** Find the 12th term of the sequence.

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STATION 13

UNIT 5 REVIEW #2

# Build an arithmetic sequence

Write an arithmetic sequence in which $a_1 = 3$ and $a_4 = 15$. Find the 13th term.

**a)** Write the sequence.

**b)** Find the 13th term.

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STATION

14

UNIT 5 REVIEW #2

# Geometric sequence with fractions

The terms of a geometric sequence are given below.

$1,\ \frac{1}{3},\ \frac{1}{9},\ \frac{1}{27}, \ldots$

**a)** Write a rule for the $n$th term of this geometric sequence.

**b)** Determine the 7th term.

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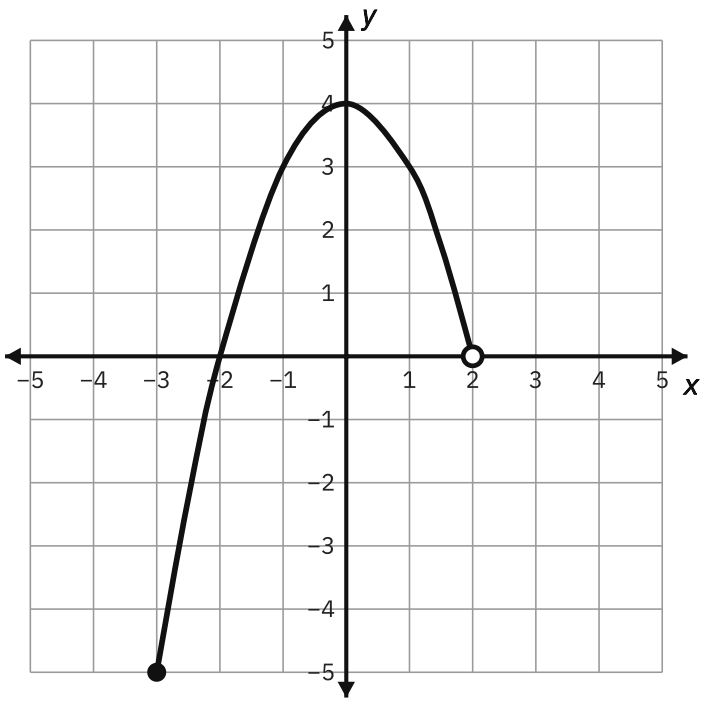
STATION

# 15

UNIT 5 REVIEW #2

## Domain and range

State the domain and range of the function.




**A)** Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 4$

**B)** Domain: $-5 \le x \le 4$
Range: $-3 \le y < 2$

**C)** Domain: $-3 \le x < 2$
Range: $-5 \le y \le 4$

**D)** Domain: $-3 \le x < 2$
Range: $-5 \le y \le 0$

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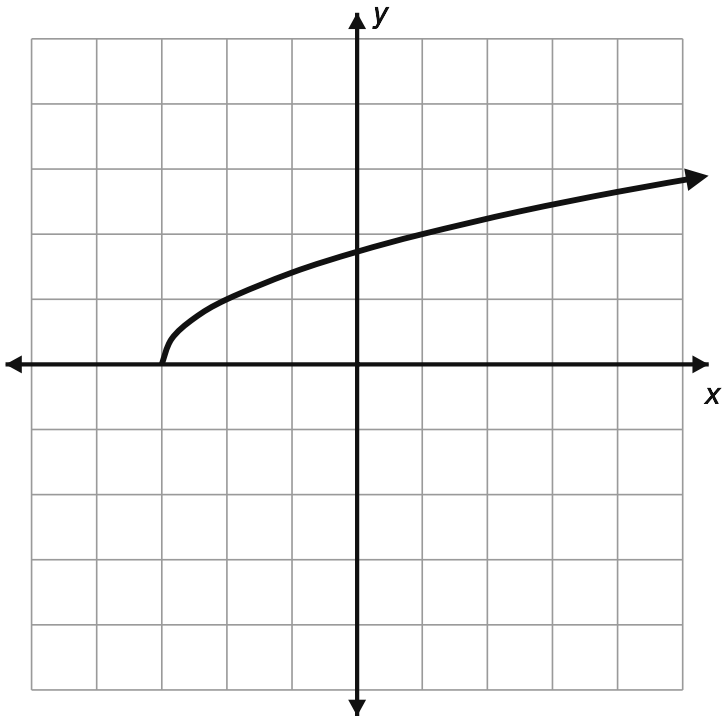
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STATION 16

UNIT 5 REVIEW #2

# Domain and range

State the domain and range of the function.




**A)** Domain: $x \geq 0$
Range: $y \geq -3$

**B)** Domain: $-3 \leq x \leq 5$
Range: $0 \leq y \leq 3$

**C)** Domain: all real numbers
Range: $y \geq 0$

**D)** Domain: $x \geq -3$
Range: $y \geq 0$

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**Another way to write it**
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**Need a hint?**
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STATION

17

UNIT 5 REVIEW #2

# Common ratio of −2

A geometric sequence with a common ratio of −2 is

**a)** 6, 4, 2, 0, ...

**b)** 5, 7, 9, 11, ...

**c)** −3, 6, −12, 24, ...

**d)** −10, −8, −6, −4, ...

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STATION

# 18

UNIT 5 REVIEW #2

## Solve for x

If $f(x) = \frac{5x - 9}{2}$, algebraically determine the value of $x$ when $f(x) = 8$.

Show every step.

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